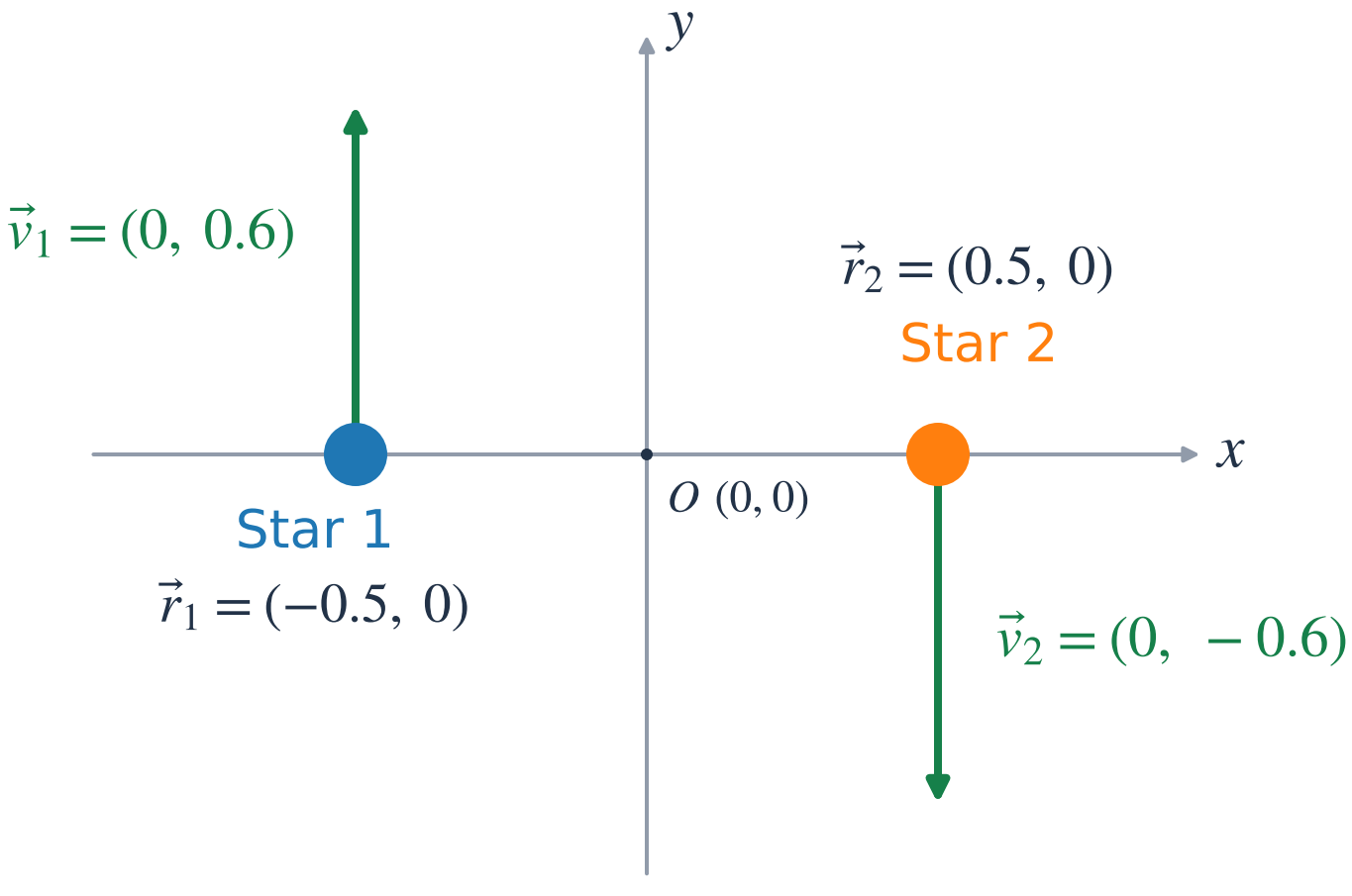
Initial positions and velocities of two stars



$Gm_1 = Gm_2 = 1$ | Equal masses; equal and opposite initial velocities.